# Tô Gia Bảo

Data Engineer

*Thu Duc, Ho Chi Minh City, Vietnam*

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## Summary

Data Engineer in digital banking with experience implementing and operating ETL pipelines using Python, SQL, and AWS. Hands-on with data lakes, dimensional modeling, PII masking, automated testing, and operational reporting. Optimized storage for an S3 table by 45% and delivered dbt models and dashboards for monitoring data platform health.

## Experience

**Timo Digital Bank by BVBank** — *Ho Chi Minh City, Vietnam*

Data Engineer — *Mar. 2025 – Present*

- Implemented MariaDB ingestion workflows with Amazon EMR and AWS Glue, applying PII masking in the curated layer and dbt transformations to deliver analytics-ready datasets.
- Optimized storage for one S3 table through Parquet ZSTD-3 compression and file compaction, saving 38.4 TB (45%) and reducing object count by 80% across 56.8 billion rows (2023 to Sep. 2026).
- Developed dbt fact and dimension models with defined grains, keys, and relationships, powering Lightdash dashboards for Airflow reliability, SLA tracking, Athena usage, and S3 storage health.
- Used dbt tests, Great Expectations, and pytest to validate data transformations and integrations, including regression checks for query contracts, SQL compilation, and dbt/Lightdash metadata consistency.
- Developed a FastMCP server powering conversational analytics in Slack, grounded in dbt docs, Lightdash dashboards, and semantic-layer metadata for self-service data discovery.
- Automated MongoDB backup validation and general ledger lookups with Python and Airflow, reducing manual checks.
- Migrated Timo Metrics from Loki/Grafana to SigNoz, centralizing logs and metrics to support operational monitoring and troubleshooting.
- Delivered Tracking247 to sync Customer-provided shipment IDs and status updates into MongoDB via Airflow and Streamlit.

## Projects

**Mini Lakehouse** — *GitHub*

Lakehouse & Platform Engineering — *Jul. 2026 – Present*

- Built a contract-driven AWS lakehouse on S3, Iceberg, Glue, Athena, and EMR Serverless, with Airflow and platform services self-hosted on OCI and infrastructure managed through Terraform.
- Delivered deterministic batch and streaming pipelines for GitHub activity, arXiv research, and SSI market data, with dbt/Lightdash analytics, Modal GPU OCR, and Netdata observability exposed through Cloudflare Zero Trust without public host ingress.

**VN Market Pulse** — *GitHub*

AI Research & Content Platform — *Jul. 2026*

- Built a Vietnamese research pipeline with multi-source search, deduplication, and reranking to generate source-backed posts.
- Used PydanticAI for typed output and validation, with retrieval budgets, retries, usage tracking, and CLI/Streamlit interfaces.

## Technical Skills

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| **Programming & Processing** | Python, SQL, Spark/PySpark, Databricks, Amazon EMR |
| **Storage & Modeling** | Amazon S3, Apache Iceberg, PostgreSQL, MariaDB, MongoDB; dimensional modeling |
| **Pipelines & Analytics** | Airflow, dbt, AWS Glue, Azure Data Factory (ADF), Amazon Athena, Lightdash |
| **Data Quality & Testing** | dbt tests, Great Expectations, pytest |
| **DevOps & Observability** | Terraform, Docker, Git, SigNoz, Netdata |

## Education

**University of Science, VNU-HCM (HCMUS)** — *Ho Chi Minh City, Vietnam*

Bachelor of Data Science | GPA: 3.78/4.00 — *2022 – 2026*

## Awards & Certifications

2025 **Top 25**, AIC Challenge
2025 **Recipient**, Ninety Eight Scholarship
2024 **Top 10**, RMIT Data Analytics Competition
2024 **Google Business Intelligence**, Professional Certificate
2024 **Google Data Analytics**, Professional Certificate

## Publication

**GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval** — *SOICT*

Co-author — *2025*

## Languages

**English** TOEIC Listening & Reading: 910 | TOEIC Speaking & Writing: 350